

Nearest Neighbor Graphs and Percolation

Asymptotic Risk, Metric Separability, and Percolation Boundaries in Spatial k -NN Lattices

A Rigorous Analytical Derivation, Stochastic Analysis, and Empirical Verification

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Abstract

This paper presents a formal analytical formulation of k -Nearest Neighbor (k -NN) graphs, metric space separability, and asymptotic risk limits.

By modeling point sets over separable metric spaces $(\mathcal{X}, D)$ with query points belonging to the support of probability measure P , we establish the almost sure convergence of nearest neighbors $X'_n \rightarrow X$ and evaluate the asymptotic Cover-Hart bound $R^* \leq R^{(1)} \leq 2R^*(1 - R^*)$ relating 1-NN risk to the optimal Bayes risk R^* .

We establish the analytical bridge linking metric space continuity to spatial lattice connectivity, examine topological phase transitions in spatial graph percolation, and evaluate a companion interactive web simulation.

My Live Simulation Engine : [Access Interactive Simulation Web Application](#)

Core Math Domain: Graph Theory & Computational Geometry

Primary Focus: Topological Invariants and Phase Transitions in Spatial k -NN Lattices

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1 Problem Formulation & Metric Space Geometry

Consider a set of n independent and identically distributed (i.i.d.) observations $\{X_1, X_2, \dots, X_n\}$ drawn from a separable metric space $(\mathcal{X}, D)$ governed by an underlying probability measure P , where $D : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ represents a valid distance metric satisfying non-negativity, symmetry, and the triangle inequality.

For a query point $X \in \text{supp}(P) \subseteq \mathcal{X}$, let $X'_n(X)$ denote its closest neighbor among the sample set $\{X_1, \dots, X_n\}$ under distance metric D .

$$X'_n(X) = \arg \min_{X_i \in \{X_1, \dots, X_n\}} D(X_i, X) \quad (1)$$

Equation (1) defines the 1-nearest neighbor mapping. Because samples are continuous random variables, spatial graph edges generated by connecting each point X to its k -nearest neighbors construct a spatial directed graph $G_k = (V, E_k)$.

1.1 Primary Investigation Objectives

The core analytical goals of this research paper are defined across two operational domains:

- **Analytical Derivation:** Prove almost sure convergence $X'_n \xrightarrow{\text{a.s.}} X$ in separable metric spaces under valid distribution support, and establish the asymptotic Cover-Hart error bounds.
- **Topological Graph Percolation:** Model spatial k -NN connectivity and evaluate giant component phase transitions as neighbor degree k scales.

2 Theoretical Proofs & Asymptotic Convergence

2.1 Almost Sure Convergence of Nearest Neighbors

We establish that as sample size $n \rightarrow \infty$, the distance between a query point X and its nearest neighbor X'_n converges to zero almost surely.

Let $S_x(\delta) = \{z \in \mathcal{X} \mid D(z, x) \leq \delta\}$ represent an open ball centered at x with radius $\delta > 0$. Crucially, because $x \in \text{supp}(P)$ lies strictly within the support of the probability measure P , every open ball $S_x(\delta)$ carries strictly positive probability measure: $\mathbb{P}\{S_x(\delta)\} = p_\delta > 0$.

The probability that no sample falls within $S_x(\delta)$ in n independent trials is given by the power probability:

$$\mathbb{P}\left(D(X'_n(x), x) > \delta\right) = \left(1 - \mathbb{P}\{S_x(\delta)\}\right)^n = (1 - p_\delta)^n \quad (2)$$

Equation (2) expresses the exact tail probability. Since $p_\delta > 0$ for all $\delta > 0$ when $x \in \text{supp}(P)$, taking the limit as $n \rightarrow \infty$ forces exponential decay to zero.

$$\lim_{n \rightarrow \infty} \mathbb{P}\left(D(X'_n(x), x) > \delta\right) = 0 \implies X'_n \xrightarrow{\text{a.s.}} X \quad (3)$$

Equation (3) completes the almost sure convergence proof via the Borel-Cantelli Lemma.

2.2 Asymptotic 1-NN Risk Function

Let $Y \in \{0, 1\}$ denote class labels, and let $\eta(X) = \mathbb{P}(Y = 1 \mid X)$ represent the conditional class probability function. Under continuous density assumptions where $\eta(x)$ is continuous over $\text{supp}(P)$, the almost sure convergence $X'_n \rightarrow X$ implies that $\eta(X'_n) \rightarrow \eta(X)$ almost surely. Thus, the sample label Y' and target label Y become conditionally independent Bernoulli variables given X .

$$R^{(1)} = \mathbb{E}\left[\mathbb{P}(Y' = 1, Y = 0 \mid X) + \mathbb{P}(Y' = 0, Y = 1 \mid X)\right] \quad (4)$$

Equation (4) formulates the asymptotic 1-NN classification risk $R^{(1)}$. Expanding conditional probabilities yields a functional expectation dependent on $\eta(X)$.

$$R^{(1)} = \mathbb{E}\left[\eta(X)(1 - \eta(X)) + (1 - \eta(X))\eta(X)\right] = \mathbb{E}\left[2\eta(X)(1 - \eta(X))\right] \quad (5)$$

Equation (5) establishes the asymptotic risk formula for the 1-NN classifier.

2.3 The Cover-Hart Theorem

Let $R^* = \mathbb{E}[\min\{\eta(X), 1 - \eta(X)\}]$ represent the optimal Bayes decision risk. Cover and Hart proved that the asymptotic 1-NN risk $R^{(1)}$ is strictly bounded by the Bayes risk.

$$R^* \leq R^{(1)} \leq 2R^*(1 - R^*) \quad (6)$$

Equation (6) defines the upper and lower Cover-Hart bounds. This demonstrates that the 1-NN classifier's asymptotic error is at most twice the optimal Bayes error rate.

3 Topological Percolation & k -NN Lattices

When spatial k -NN graphs are constructed over random point fields in $\mathbb{R}^2$, the graph topology exhibits a fundamental phase transition in component connectivity as degree parameter k increases.

3.1 Bridge: Graph Connectivity and Risk Consistency

Spatial k -NN connectivity serves as the structural foundation for local metric consistency. For k -NN classifiers to achieve asymptotic risk consistency ($\eta(X'_n) \rightarrow \eta(X)$), query points must connect to local neighbors within a connected spatial neighborhood. As degree k scales, spatial graph edges construct a connected lattice that spans local probability manifolds.

3.2 Phase Transitions in Spatial Graph Connectivity

For small neighbor limits $k < k_c$, the spatial graph G_k consists of small, disconnected clusters and isolated cycles. As k crosses a critical threshold k_c , a giant connected component emerges across the spatial domain.

$$k_c = \arg \min_k \left\{ \lim_{N \rightarrow \infty} \frac{|C_{\max}|}{N} > 0 \right\} \quad (7)$$

Equation (7) formalizes the percolation threshold condition, where $|C_{\max}|$ represents the order of the largest connected subgraph.

- **Sub-Critical Regime** ($k < k_c$): Graph contains isolated cluster pockets; distance matrices show fragmented components.
- **Percolation Boundary** ($k = k_c$): A giant connected component forms, exhibiting topological phase transitions.
- **Super-Critical Regime** ($k > k_c$): Graph achieves global topological connectivity across all spatial nodes.

4 Applications in Machine Learning & Computational Geometry

Spatial k -NN graph structures and metric convergence properties serve as core foundations across modern computational intelligence:

4.1 Manifold Learning & Dimensionality Reduction

Algorithms like ISOMAP and t-SNE construct k -NN spatial lattices to approximate non-linear geodesic manifold distances prior to low-dimensional embedding.

4.2 Non-Parametric Clustering & Anomaly Detection

Density-based clustering methods (e.g., DBSCAN) and local outlier factor (LOF) models rely on spatial nearest-neighbor distances $D(X'_k, X)$ to estimate local probability density gradients.

5 Interactive Web Simulation Evaluation

To complement this theoretical monograph, an interactive single-file web application was developed: `nearest_neighbor_sim.html`.

- **Live Interactive Application:** <https://nearestneighborgraphs.vercel.app/>
- **GitHub Source Code Repository:** <https://github.com/wheyyydant/math-drafts-and-si>
- **2D Node Generation Canvas:** Interactive HTML5 Canvas rendering 2D spatial point fields, dynamically illustrating k -NN edge allocation and cluster formation.
- **Percolation Tracker:** Real-time component tracking and distance matrix calculations visualizing giant component phase transitions as k scales.

6 Concluding Remarks

We have presented an analytical derivation of k -NN graph asymptotic convergence and Cover-Hart error bounds $R^{(1)} \leq 2R^*(1 - R^*)$ over separable metric spaces. Spatial k -NN graph simulations confirm giant component percolation phase transitions as degree parameter k scales.

7 Mathematical Glossary

Separable Metric Space:	A metric space containing a countable dense subset, ensuring almost sure convergence of nearest neighbors under support conditions.
Probability Measure Support ($\text{supp}(P)$):	The closure of the set of points where every open neighborhood carries strictly positive probability measure.
Bayes Risk (R^*):	The minimum achievable classification error rate under optimal decision boundaries.
Cover-Hart Bound:	The theoretical limit proving $R^{(1)} \leq 2R^*(1 - R^*)$ for 1-NN classifiers.
Percolation Threshold (k_c):	The minimum degree k at which a giant connected component spans a spatial random graph.